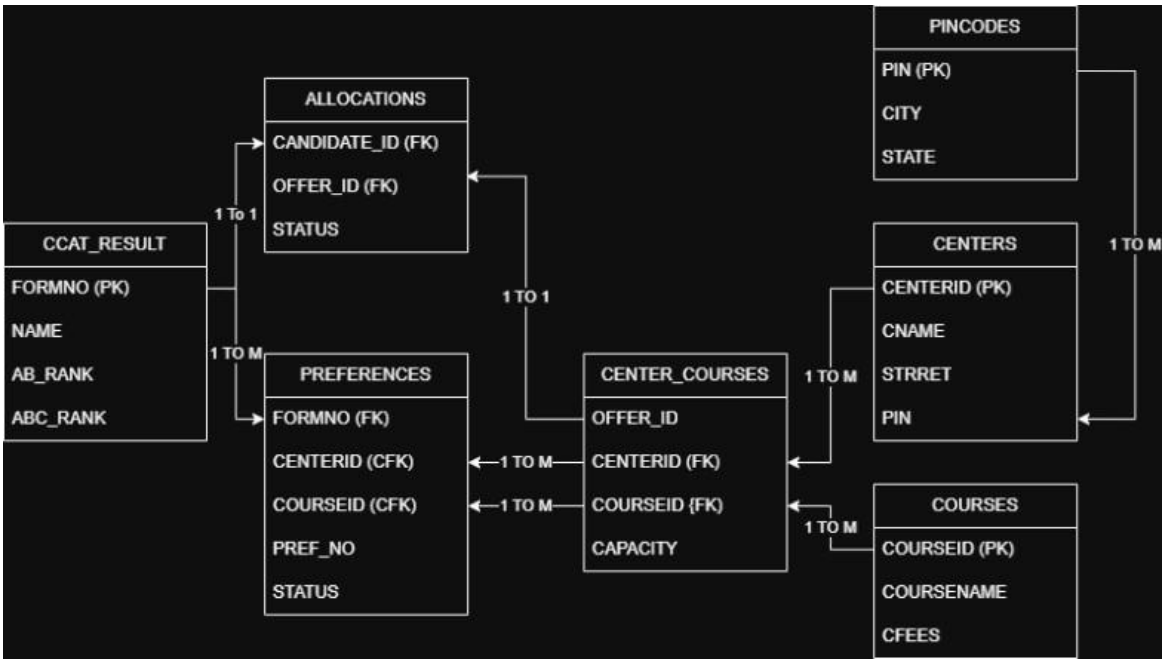
CDAC ADMISSION DATABASE

- **Design database for CDAC Admission System.**
  Create appropriate tables and insert sample records.
  Write a Stored Procedure for Round1 allocation.
  Write a Stored Procedure for Round2 allocation.

ER DIAGRAM:



RELATIONS:

Entity 1	Relationship	Entity 2	Type
Pincode	contains	Center	1 : M
Center	offers	Center_Course	1 : M
Course	included in	Center_Course	1 : M
Center_Course	listed in	Preference	1 : M
Candidate	gives	Preference	1 : M
Center_Course	assigned via	Allocation	1 : 1
Candidate	receives	Allocation	1 : 1

CREATING DATABASE:

```
root> create database cdac_admission;
Query OK, 1 row affected (0.167 sec)

root> grant all on cdac_admission.* to D2_96718_harish@localhost
    -> ;
Query OK, 0 rows affected (0.135 sec)

root> SET GLOBAL log_bin_trust_function_creators = 1;
Query OK, 0 rows affected, 1 warning (0.006 sec)
```

```
Database changed
d2_96695_sahil> CREATE TABLE cdac_candidate (
    ->     candidate_id   INT AUTO_INCREMENT PRIMARY KEY,
    ->     application_no VARCHAR(20) NOT NULL UNIQUE,
    ->     candidate_name VARCHAR(100) NOT NULL,
    ->     email          VARCHAR(100) NOT NULL UNIQUE,
    ->     merit_rank     INT NOT NULL UNIQUE,
    ->     entrance_score DECIMAL(5,2) NOT NULL,
    ->     category       ENUM('OPEN','OBC','SC','ST','EWS','PwD') NOT NULL DEFAULT 'OPEN',
    ->     gender         ENUM('M','F','O') NOT NULL DEFAULT 'O',
    ->     status         ENUM('NEW','ALLOCATED','WAITLISTED') NOT NULL DEFAULT 'NEW',
    ->     created_at     TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     CHECK (merit_rank > 0),
    ->     CHECK (entrance_score BETWEEN 0 AND 100)
    -> );
Query OK, 0 rows affected (0.13 sec)

d2_96695_sahil>
```

```
d2_96695_sahil> CREATE TABLE cdac_program (
    ->     program_id    INT AUTO_INCREMENT PRIMARY KEY,
    ->     program_code  VARCHAR(20) NOT NULL UNIQUE,
    ->     program_name  VARCHAR(100) NOT NULL,
    ->     center_name   VARCHAR(100) NOT NULL,
    ->     round1_seats  INT NOT NULL DEFAULT 0,
    ->     round2_seats  INT NOT NULL DEFAULT 0,
    ->     CHECK (round1_seats >= 0),
    ->     CHECK (round2_seats >= 0),
    ->     UNIQUE (program_name, center_name)
    -> );
Query OK, 0 rows affected (0.05 sec)

d2_96695_sahil>
```

```
d2_96695_sahil> CREATE TABLE cdac_preference (
    ->     candidate_id INT NOT NULL,
    ->     pref_no      INT NOT NULL,
    ->     program_id   INT NOT NULL,
    ->     PRIMARY KEY (candidate_id, pref_no),
    ->     UNIQUE (candidate_id, program_id),
    ->     FOREIGN KEY (candidate_id) REFERENCES cdac_candidate(candidate_id
)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->     FOREIGN KEY (program_id) REFERENCES cdac_program(program_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->     CHECK (pref_no BETWEEN 1 AND 5)
    -> );
Query OK, 0 rows affected (0.08 sec)

d2_96695_sahil>
```

```
d2_96695_sahil> CREATE TABLE cdac_allocation (
    ->     allocation_id     INT AUTO_INCREMENT PRIMARY KEY,
    ->     academic_year     VARCHAR(9) NOT NULL,
    ->     candidate_id      INT NOT NULL UNIQUE,
    ->     program_id        INT NOT NULL,
    ->     round_no          INT NOT NULL,
    ->     allocation_status ENUM('ALLOTTED','REPORTED','CANCELLED') NOT NUL
L DEFAULT 'ALLOTTED',
    ->     allotted_on       TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (candidate_id) REFERENCES cdac_candidate(candidate_id
)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->     FOREIGN KEY (program_id) REFERENCES cdac_program(program_id)
    ->         ON DELETE CASCADE ON UPDATE CASCADE,
    ->     CHECK (round_no IN (1,2))
    -> );
Query OK, 0 rows affected (0.07 sec)

d2_96695_sahil> |
```

SAMPLE DATA INSERTION:

```
d2_96695_sahil> INSERT INTO cdac_program (program_code, program_name, center_name, round1_seats, round2_seats) VALUES
    -> ('PG-DAC',   'Diploma in Advanced Computing',     'Pune',       1, 1),
    -> ('PG-DBDA',  'Big Data Analytics',                'Mumbai',     1, 1),
    -> ('PG-DITISS','IT Infrastructure, Systems & Security', 'Pune',   1, 0),
    -> ('PG-DSDA',  'Data Science & AI',                 'Bangalore',  0, 1);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

d2_96695_sahil> |
```

```
d2_96695_sahil> INSERT INTO cdac_candidate
    -> (application_no, candidate_name, email, merit_rank, entrance_score, category, gender)
    -> VALUES
    -> ('APP1001', 'Rahul Sharma', 'rahul@example.com', 1, 98.50, 'OPEN', 'M'),
    -> ('APP1002', 'Neha Patil',   'neha@example.com',  2, 97.00, 'OBC',  'F'),
    -> ('APP1003', 'Amit Kumar',   'amit@example.com',  3, 95.20, 'OPEN', 'M'),
    -> ('APP1004', 'Priya Desai',  'priya@example.com', 4, 93.10, 'SC',   'F'),
    -> ('APP1005', 'Karan Joshi',  'karan@example.com', 5, 91.40, 'OPEN', 'M'),
    -> ('APP1006', 'Isha Verma',   'isha@example.com',  6, 90.00, 'EWS',   'F');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

d2_96695_sahil> |
```

```
d2_96695_sahil>
d2_96695_sahil> INSERT INTO cdac_preference VALUES
    -> (1, 1, 1),
    -> (1, 2, 2),
    -> (1, 3, 3);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

d2_96695_sahil> |
```

```
d2_96695_sahil> INSERT INTO cdac_preference VALUES
    -> (2, 1, 2),
    -> (2, 2, 1),
    -> (2, 3, 3);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

d2_96695_sahil>
```

```
d2_96695_sahil> INSERT INTO cdac_preference VALUES
    -> (3, 1, 3),
    -> (3, 2, 1),
    -> (3, 3, 2);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

d2_96695_sahil>
```

```
d2_96695_sahil> INSERT INTO cdac_preference VALUES
    -> (4, 1, 1),
    -> (4, 2, 4),
    -> (4, 3, 2);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

d2_96695_sahil>
```

```
d2_96695_sahil> INSERT INTO cdac_preference VALUES
    -> (5, 1, 4),
    -> (5, 2, 2),
    -> (5, 3, 1);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

d2_96695_sahil>
```

```
d2_96695_sahil> INSERT INTO cdac_preference VALUES
    -> (6, 1, 2),
    -> (6, 2, 4),
    -> (6, 3, 3);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

d2_96695_sahil>
```

CREATE PROCEDURE FOR ROUND ALLOCATION:

```
d2_96695_sahil> DELIMITER $$
d2_96695_sahil>
d2_96695_sahil> CREATE PROCEDURE sp_allocate_round(
    ->     IN p_academic_year VARCHAR(9),
    ->     IN p_round_no INT,
    ->     OUT p_allocated INT
    -> )
    -> BEGIN
    ->     DECLARE done INT DEFAULT 0;
    ->     DECLARE v_candidate_id INT;
    ->     DECLARE v_pref_no INT;
    ->     DECLARE v_program_id INT;
    ->     DECLARE v_seats_left INT;
    ->     DECLARE v_allocated_flag INT DEFAULT 0;
    ->     DECLARE v_max_pref INT DEFAULT 5;
    ->
    ->     DECLARE cur_candidates CURSOR FOR
    ->         SELECT c.candidate_id
    ->         FROM cdac_candidate c
    ->         WHERE NOT EXISTS (
    ->             SELECT 1
    ->             FROM cdac_allocation a
    ->             WHERE a.candidate_id = c.candidate_id
    ->               AND a.academic_year = p_academic_year
    ->               AND a.allocation_status = 'ALLOTTED'
    ->         )
    ->         ORDER BY c.merit_rank, c.candidate_id;
    ->
    ->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
    ->
    ->     DECLARE EXIT HANDLER FOR SQLEXCEPTION
    ->     BEGIN
    ->         ROLLBACK;
    ->         RESIGNAL;
    ->     END;
    ->
    ->     SET p_allocated = 0;
    ->
    ->     START TRANSACTION;
    ->
    ->     OPEN cur_candidates;
    ->
    ->     read_loop: LOOP
    ->         FETCH cur_candidates INTO v_candidate_id;
    ->         IF done = 1 THEN
    ->             LEAVE read_loop;
    ->         END IF;
    ->
    ->         SET v_pref_no = 1;
    ->         SET v_allocated_flag = 0;
    ->
    ->         pref_loop: WHILE v_pref_no <= v_max_pref AND v_allocated_flag = 0 DO
    ->             SET v_program_id = NULL;
    ->
    ->             BEGIN
    ->                 DECLARE v_missing INT DEFAULT 0;
    ->                 DECLARE CONTINUE HANDLER FOR NOT FOUND SET v_missing = 1;
    ->
    ->                 SELECT p.program_id
    ->                 INTO v_program_id
    ->                 FROM cdac_preference p
    ->                 WHERE p.candidate_id = v_candidate_id
    ->                   AND p.pref_no = v_pref_no
    ->                 LIMIT 1;
    ->             END;
    ->
    ->             IF v_program_id IS NOT NULL THEN
    ->                 SELECT
    ->                     CASE
    ->                         WHEN p_round_no = 1 THEN
    ->                             pr.round1_seats - COALESCE((
    ->                                 SELECT COUNT(*)
    ->                                 FROM cdac_allocation a
    ->                                 WHERE a.program_id = pr.program_id
    ->                                   AND a.academic_year = p_academic_year
    ->                                   AND a.allocation_status = 'ALLOTTED'
    ->                             ), 0)
    ->                         ELSE
    ->                             (pr.round1_seats + pr.round2_seats) - COALESCE((
    ->                                 SELECT COUNT(*)
    ->                                 FROM cdac_allocation a
    ->                                 WHERE a.program_id = pr.program_id
    ->                                   AND a.academic_year = p_academic_year
    ->                                   AND a.allocation_status = 'ALLOTTED'
    ->                             ), 0)
    ->                     END
    ->                 INTO v_seats_left
    ->                 FROM cdac_program pr
    ->                 WHERE pr.program_id = v_program_id;
    ->
    ->                 IF v_seats_left > 0 THEN
    ->                     INSERT INTO cdac_allocation (
    ->                         academic_year, candidate_id, program_id, round_no
    ->                     )
    ->                     VALUES (
    ->                         p_academic_year, v_candidate_id, v_program_id, p_round_no
    ->                     );
    ->
    ->                     UPDATE cdac_candidate
    ->                     SET status = 'ALLOCATED'
    ->                     WHERE candidate_id = v_candidate_id;
    ->
    ->                     SET p_allocated = p_allocated + 1;
    ->                     SET v_allocated_flag = 1;
    ->                 END IF;
    ->             END IF;
    ->
    ->             SET v_pref_no = v_pref_no + 1;
    ->         END WHILE;
    ->     END LOOP;
    ->
    ->     CLOSE cur_candidates;
    ->
    ->     COMMIT;
    -> END$$
Query OK, 0 rows affected (0.03 sec)

d2_96695_sahil>
d2_96695_sahil> DELIMITER ;
```

PROCEDURE FOR ROUND ALLOCATIONS:

```
d2_96695_sahil> DELIMITER $$
d2_96695_sahil>
d2_96695_sahil> CREATE PROCEDURE sp_round1_allocation(
    ->     IN p_academic_year VARCHAR(9),
    ->     OUT p_allocated INT
    -> )
    -> BEGIN
    ->     CALL sp_allocate_round(p_academic_year, 1, p_allocated);
    -> END$$
Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil>
d2_96695_sahil> CREATE PROCEDURE sp_round2_allocation(
    ->     IN p_academic_year VARCHAR(9),
    ->     OUT p_allocated INT
    -> )
    -> BEGIN
    ->     CALL sp_allocate_round(p_academic_year, 2, p_allocated);
    -> END$$
Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil>
d2_96695_sahil> DELIMITER ;
d2_96695_sahil>
```

RESULTS:

```
d2_96695_sahil> CALL sp_round1_allocation('2025-26', @r1);
Query OK, 0 rows affected (0.04 sec)

d2_96695_sahil> SELECT @r1 AS round1_allocated;
+------------------+
| round1_allocated |
+------------------+
|                3 |
+------------------+
1 row in set (0.00 sec)

d2_96695_sahil>
d2_96695_sahil> CALL sp_round2_allocation('2025-26', @r2);
Query OK, 0 rows affected (0.01 sec)

d2_96695_sahil> SELECT @r2 AS round2_allocated;
+------------------+
| round2_allocated |
+------------------+
|                3 |
+------------------+
1 row in set (0.00 sec)

d2_96695_sahil>
d2_96695_sahil> SELECT * FROM cdac_allocation ORDER BY round_no, candidate_id;
+---------------+---------------+--------------+------------+----------+------------------+---------------------+
| allocation_id | academic_year | candidate_id | program_id | round_no | allocation_status | allotted_on         |
+---------------+---------------+--------------+------------+----------+------------------+---------------------+
|             1 | 2025-26       |            1 |          1 |        1 | ALLOTTED         | 2026-03-19 21:56:00 |
|             2 | 2025-26       |            2 |          2 |        1 | ALLOTTED         | 2026-03-19 21:56:00 |
|             3 | 2025-26       |            3 |          3 |        1 | ALLOTTED         | 2026-03-19 21:56:00 |
|             4 | 2025-26       |            4 |          1 |        2 | ALLOTTED         | 2026-03-19 21:56:00 |
|             5 | 2025-26       |            5 |          4 |        2 | ALLOTTED         | 2026-03-19 21:56:00 |
|             6 | 2025-26       |            6 |          2 |        2 | ALLOTTED         | 2026-03-19 21:56:00 |
+---------------+---------------+--------------+------------+----------+------------------+---------------------+
6 rows in set (0.00 sec)

d2_96695_sahil> |
```